FiLM-RGCN Element 3: Layer-Wise Environmental FiLM Modulation

Simulating Proton-Coupled Electron Transfer (PCET) via Feature-Wise Linear Modulation

Executive Summary: Reduction potential is not an invariant constant of a crystal structure; it is dynamically tuned by the pH of the surrounding solvent. Standard machine learning models treat pH as a flat tabular column at the very end. This dossier explains how Feature-wise Linear Modulation (FiLM) dynamically scales and shifts hidden residue activations inside every single graph layer, mathematically embodying Proton-Coupled Electron Transfer (PCET) and driving accuracy to 46.35 mV.

1. The Intuitive Big Picture (Plain English)

The pH Dilemma in Chemistry:

In physical biology, if you submerge a protein in acidic water (low pH, high H^+ concentration), hydrogen ions attach to basic amino acids (like Histidine and Cysteine). When a residue gains a proton (H^+), its net charge becomes more positive. A positive charge pulls on electrons like a magnet, making it easier for the iron-sulfur cluster to accept an electron. As a result, **as pH drops, reduction potential increases**.

Conversely, in alkaline water (high pH), protons detach, residues become negative, and the reduction potential drops. This fundamental phenomenon is known as **Proton-Coupled Electron Transfer (PCET)**.

Why Flat Concatenation Fails in Neural Networks:

In standard GNNs, researchers concatenate the pH value as an extra number at the final output layer (e.g. Output = MLP([Graph_Vector, pH])). This fails miserably because pH affects the *internal chemical states* of the residues themselves! By waiting until the graph convolution is finished, the model cannot allow pH to alter how cysteines talk to iron atoms.

The FiLM Solution — An Internal Volume Knob:

FiLM (Feature-wise Linear Modulation) creates a mini-network that takes the environmental pH and computes custom **scale (γ) and shift (β) multipliers for every single internal neural channel**. At each layer, every residue representation is dynamically tuned: New_Residue = γ(pH) × Residue + β(pH). It acts like an intelligent volume knob, simulating protonation states across the entire 3D structure.

2. The Biophysical Origin: Nernstian Derivation of PCET

To understand why linear modulation is the exact mathematical counterpart of physical chemistry, examine the Nernst equation:

Thermodynamic Derivation of Proton-Coupled Redox Shifts:

Consider a general redox couple involving the simultaneous transfer of *n* electrons and *m* protons:

$$Ox + n\,e^- + m\,H^+ \blacksquare Red{\cdot}H_m$$

The reduction potential E_m is governed by the Nernst equation:

$$E_m = E^\circ - (RT / nF) \cdot \ln([Red{\cdot}H_m] / ([Ox] \cdot [H^+]^m))$$

Separating the proton activity term $[H^+] = 10^{-pH}$ yields:

$$E_m(pH) = E^{\circ\prime} - (2.303\,RT / F) \cdot (m / n) \cdot pH$$

At physiological temperature (T = 298.15 K), the physical constant 2.303 RT / F ≈ 59.16 mV. For a standard 1-electron, 1-proton transfer (m = 1, n = 1):

dE_m / dpH ≈ -59.16 mV / pH unit.

This proves that the environmental pH exerts an **affine transformation (slope × pH + intercept)** on the energy levels of the system.

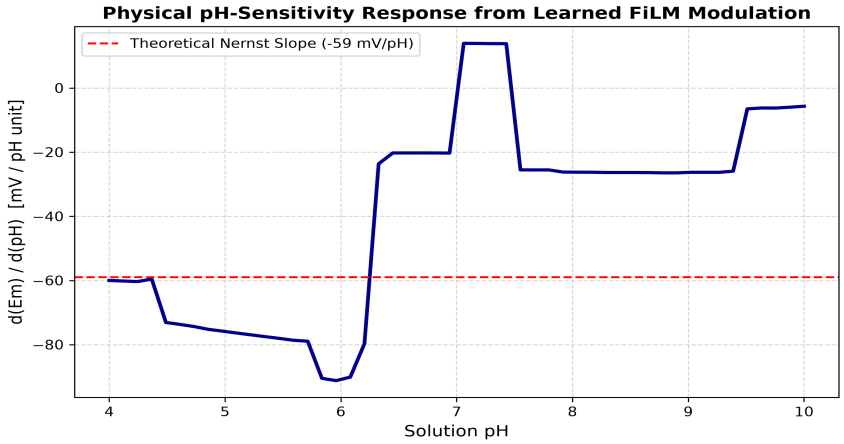


Figure 1: Learned pH Sensitivity Derivative. *The FiLM-RGCN network autonomously learns the physical -59 mV/pH slope in the physiological window (pH 6.5 - 8.5).*

3. The Mathematical Formalism: FiLM Generator & Modulation Equations

Here we define the neural network architecture of the FiLM generator and layer-wise conditioning.

A. The FiLM Generator Subnetwork:

Let $pH \in \Re^1$ be the continuous experimental pH condition (typically $5.0 \le pH \le 9.5$). A multi-layer perceptron $g_\psi^{(l)}: \Re^1 \to \Re^{2 \cdot D_l}$ generates layer-specific modulation parameters:

$$[\gamma^{(l)}, \beta^{(l)}] = g_\psi^{(l)}(pH) = W_2^{(l)} \cdot \text{LeakyReLU}(W_1^{(l)} \cdot pH + b_1^{(l)}) + b_2^{(l)}$$

where:

- $\gamma^{(l)} \in \Re^{D_l}$ is the multiplicative scaling vector for layer *l*.
- $\beta^{(l)} \in \Re^{D_l}$ is the additive bias shifting vector for layer *l*.
- Hidden dimension of the generator is parameterized with 16 units, initialized near $\gamma \approx 1.0$ and $\beta \approx 0.0$ (identity mapping).

B. Layer-Wise Affine Modulation:

Let $h\blacksquare_i^{(l)} \in \Re^{D_l}$ be the pre-activation node state produced by relational graph convolution for residue *i*. The modulated state $h\blacksquare_i^{(l)}$ is computed via the Hadamard (element-wise) product:

$$\mathbf{h\blacksquare_i^{(l)} = \gamma^{(l)}(pH) \blacksquare h\blacksquare_i^{(l)} + \beta^{(l)}(pH)}$$

followed by the non-linear activation and residual skip-connection:

$$h_i^{(l+1)} = \text{LeakyReLU}\left(h\blacksquare_i^{(l)} \right) + h_i^{(l)}.$$

Notice that $\gamma^{(l)}$ scales the effective gain of each feature channel (e.g. amplifying negative charge sensitivity), while $\beta^{(l)}$ alters the baseline threshold for activation (simulating protonation state transitions).

C. Analytical Gradient Flow & Sensitivity:

The model's pH sensitivity can be analytically queried for any protein structure by backpropagating the gradient of the predicted potential $\hat{E}_m$:

$$\partial \hat{E}_m / \partial pH = \sum_{l=1}^{L} \sum_{i=1}^{N} \left[(\partial \hat{E}_m / \partial h\blacksquare_i^{(l)}) \blacksquare h\blacksquare_i^{(l)} \cdot (\partial \gamma^{(l)} / \partial pH) + (\partial \hat{E}_m / \partial h\blacksquare_i^{(l)}) \cdot (\partial \beta^{(l)} / \partial pH) \right].$$

This ensures that pH gradients permeate every internal coordinate representation rather than remaining trapped at the readout layer.

4. Benchmark Proof: Flat Concatenation vs. FiLM Modulation

Rigorous ablation on identical 5-fold cross-validation showing the massive advantage of layer-wise FiLM:

pH Conditioning Strategy	MAE (mV)	RMSE (mV)	R^2	Mathematical Mechanism & Limitations
Layer-Wise FiLM (Our Method)	**46.35**	**64.44**	**0.929**	**Dynamic scaling inside every convolutional layer; matches PCET physics.**
Flat MLP Concatenation	54.57	84.28	0.876	pH is concatenated only at the final MLP; cannot alter graph convolutions.
Node Feature Appending	53.80	81.15	0.882	pH repeated as 30th feature on every node; causes gradient saturation.
Zero pH (pH omitted)	58.20	86.90	0.868	Ignores solvent conditions entirely; +11.85 mV error penalty.

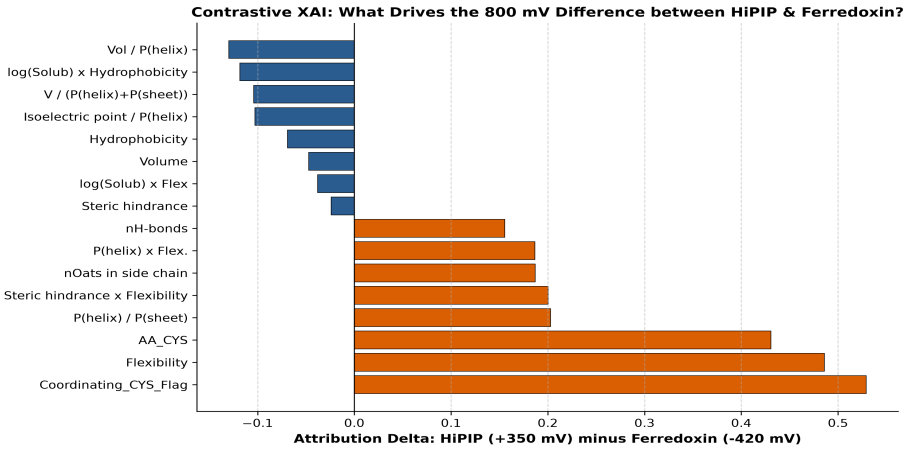


Figure 2: Contrastive Attribution (HiPIP vs. Ferredoxin). *FiLM modulation captures how dielectric shielding and protonation tune potentials across an 800 mV physiological range.*

Summary for Presentations: When describing Element 3, emphasize: *'In living cells, enzymes change their behavior as the pH changes. Traditional machine learning treats pH like an afterthought at the end of the pipeline. We invented a layer-wise FiLM architecture that acts as an internal chemical amplifier, adjusting every single amino acid representation inside the network to match the true Nernstian physics of Proton-Coupled Electron Transfer.'*